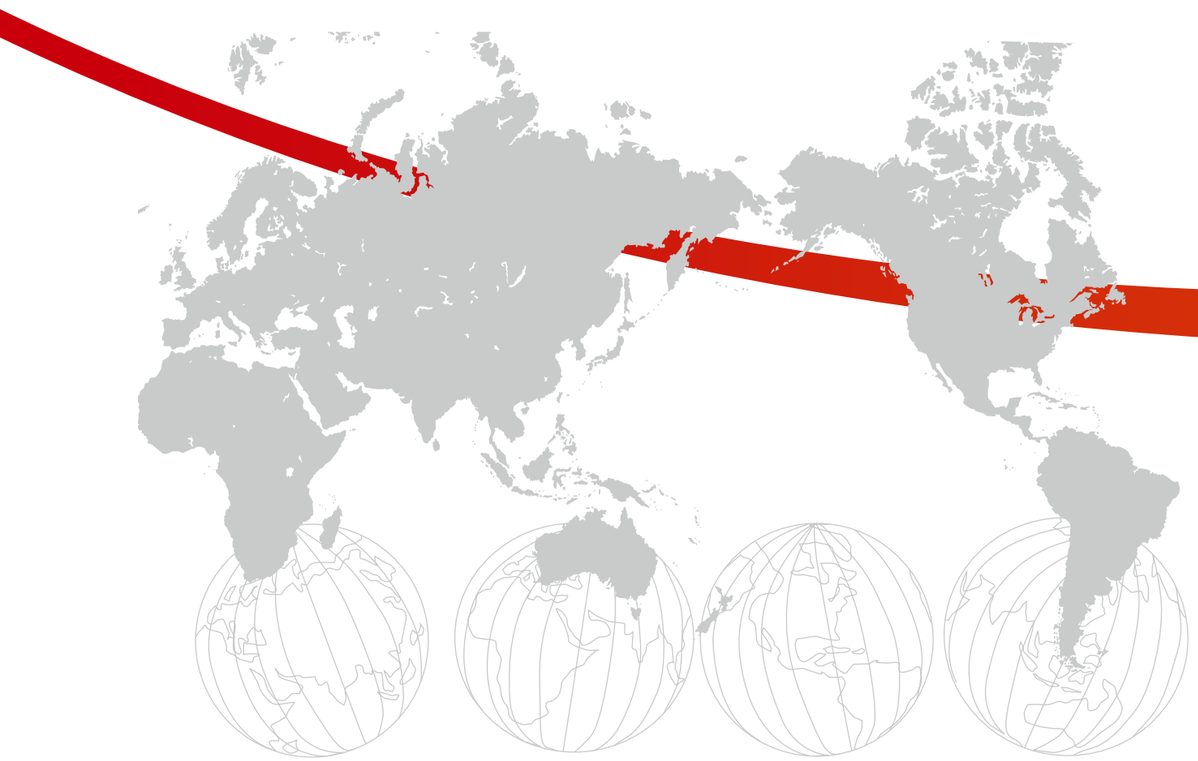


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Zhafir **Venus** II/p Series

ZHAFIR
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Zhafir **Venus** 2• Series

Packaging / International

1500-4500kN



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Technical data **VE1500 Ⅱ /p**

CLAMPING UNIT		VE1500 Ⅱ
Clamping force	kN	1500
Toggle stroke	mm	420
Min. mold height	mm	180
Max. mold height	mm	520
Max. mold opening stroke	mm	940
Space between tie bars H×V	mm	520×470
Min. mold dimension	mm	335×305
Ejector stroke	mm	120
Ejector force	kN (tf)	34.3 (3.5)
Mold platen dimension H×V	mm	770×730

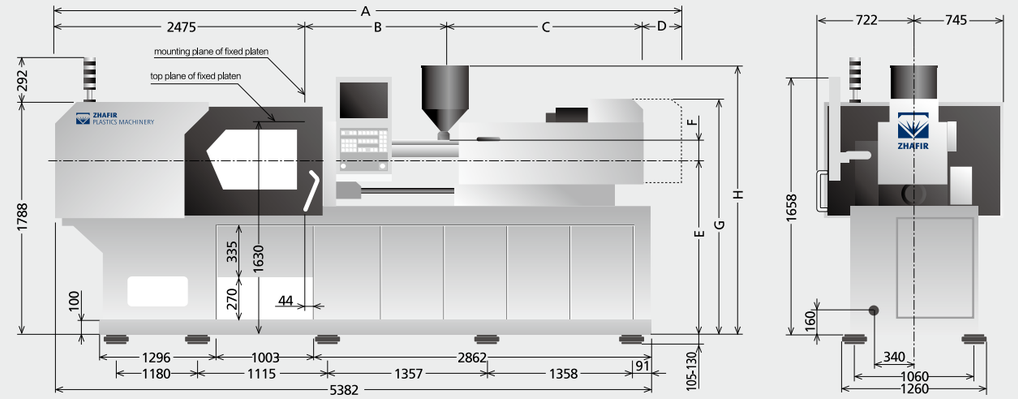
		A	B	C (OP)	A	B	C (OP)
Screw diameter	mm	36	40	45	40	45	50
Screw L/D ratio	L/D	25	22.5	23	25	22.2	23
Injection volume (theoretical)	cm^3	173	213	270	252	319	395
Injection weight (PS)	g	157	194	246	229	290	358
Screw speed	rpm		400			350	
Plasticizing capacity (PS)	g/s	22	30	42	27	39	50
Nozzle contact force	kN (tf)		29.4 (3.0)			29.4 (3.0)	
Heating power	kW	13.5	13.5	14.8	15.8	15.8	17.8
INJECTION UNIT			**430pns**			**640pns**	
Injection speed	mm/s		200			160	
Injection rate (PS)	g/s	184	228	289	183	231	285
Injection pressure	Mpa	247	200	158	253	200	162
	bar	2470	2000	1580	2530	2000	1620
INJECTION UNIT			**430p**			**640p**	
Injection speed	mm/s		300			300	
Injection rate (PS)	g/s	277	342	434	342	434	536
Injection pressure	Mpa	247	200	158	253	200	162
	bar	2470	2000	1580	2530	2000	1620

OTHERS			
Connection power	kW/A	430pns:32.3/57 430p:44.2/79	640pns:39/69 640p:61/110
Hopper capacity	L	25	25
Machine dimension	m	5.53×1.47×2.08	5.83×1.47×2.08
Machine weight	t	7.8	8.5

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE1500 Ⅱ /p

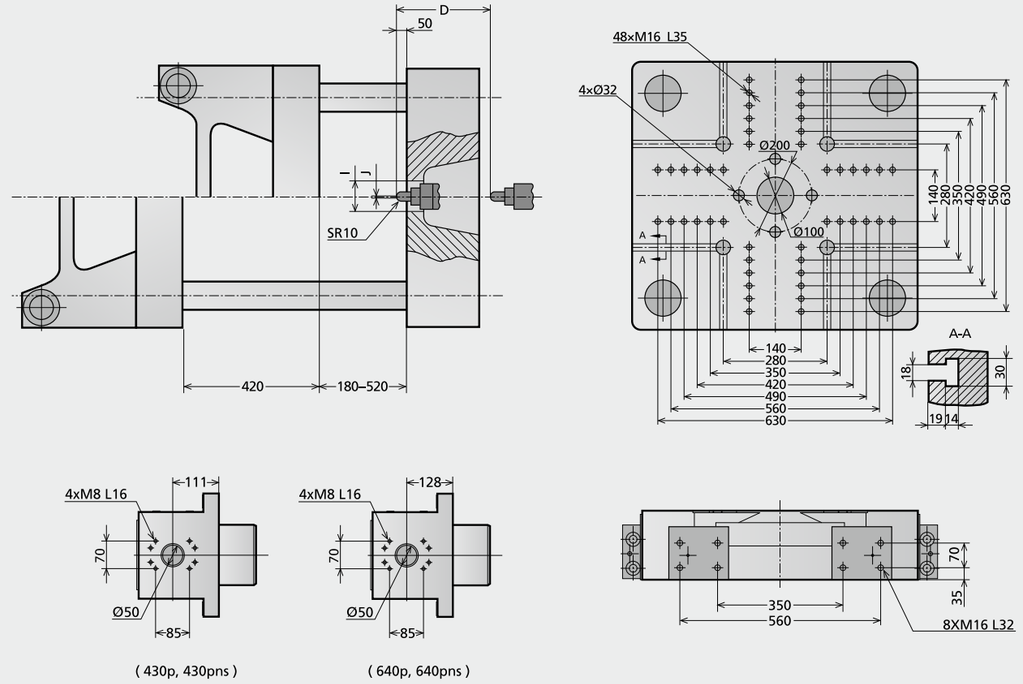
unit: mm



Note: The above machine size does not apply to the “C” type screw.

Platen dimensions VE1500 Ⅱ /p

unit: mm



(430p, 430pns)

(640p, 640pns)

	A	B	C	D	E	F	G	H	I	J
640p, 640pns	5820	1153	1822	370	1260	209	1818	1993	Ø125	Ø3
430p,430pns	5597	1044	1628	450	1260	207	1810	1990	Ø125	Ø3

Technical data **VE1900 Ⅱ /p**

CLAMPING UNIT		VE1900 Ⅱ
Clamping force	kN	1900
Toggle stroke	mm	470
Min. mold height	mm	200
Max. mold height	mm	550
Max. mold opening stroke	mm	1020
Space between tie bars H×V	mm	570×520
Min. mold dimension	mm	370×335
Ejector stroke	mm	130
Ejector force	kN (tf)	44.1 (4.5)
Mold platen dimension H×V	mm	840×790

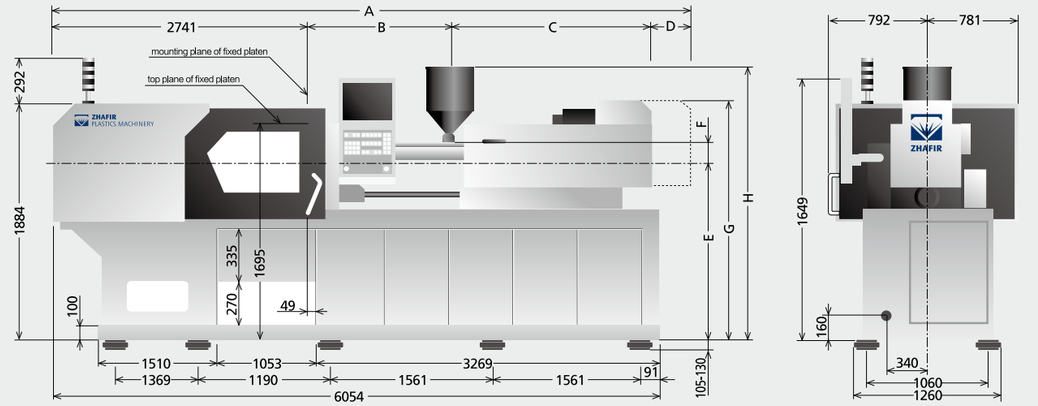
		A	B	C (OP)	A	B	C (OP)
Screw diameter	mm	40	45	50	45	50	55
Screw L/D ratio	L/D	25	22.2	23	25	22.5	23
Injection volume (theoretical)	cm^3	252	319	395	334	412	499
Injection weight (PS)	g	229	290	358	304	375	454
Screw speed	rpm		350			320	
Plasticizing capacity (PS)	g/s	27	39	50	35	46	60
Nozzle contact force	kN (tf)		39.2 (4.0)			39.2 (4.0)	
Heating power	kW	15.8	15.8	17.8	21.2	21.2	23.4
INJECTION UNIT			**640pns**			**830pns**	
Injection speed	mm/s		160			160	
Injection rate (PS)	g/s	183	231	285	231	285	346
Injection pressure	Mpa	253	200	162	247	200	165
	bar	2530	2000	1620	2470	2000	1650
INJECTION UNIT			**640p**			**830p**	
Injection speed	mm/s		300			300	
Injection rate (PS)	g/s	342	434	536	434	536	648
Injection pressure	Mpa	253	200	162	247	200	165
	bar	2530	2000	1620	2470	2000	1650

OTHERS			
Connection power	kW/A	640pns:39/69 640p:61/110	830pns:47.5/85 830p:68.6/123
Hopper capacity	L	25	50
Machine dimension	m	6.14×1.57×2.18	6.47×1.57×2.18
Machine weight	t	9.6	10

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE1900 Ⅱ /p

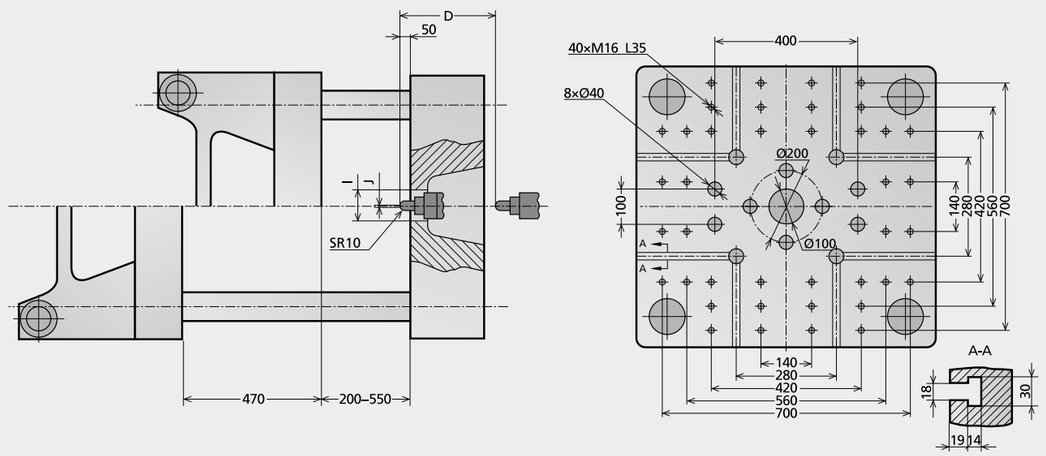
unit: mm



Note: The above machine size does not apply to the "C" type screw.

Platen dimensions VE1900 Ⅱ /p

unit: mm



(640p, 640pns)

(830p, 830pns)

	A	B	C	D	E	F	G	H	I	J
830p, 830pns	6464	1277	1996	450	1295	233	1908	2209	Ø160	Ø3
640p, 640pns	6126	1153	1822	410	1295	209	1852	2028	Ø160	Ø3

Technical data **VE2300 Ⅱ /p**

CLAMPING UNIT		VE2300 Ⅱ
Clamping force	kN	2300
Toggle stroke	mm	550
Min. mold height	mm	220
Max. mold height	mm	600
Max. mold opening stroke	mm	1150
Space between tie bars H×V	mm	620×570
Min. mold dimension	mm	400×370
Ejector stroke	mm	150
Ejector force	kN (tf)	49 (5.0)
Mold platen dimension H×V	mm	930×880

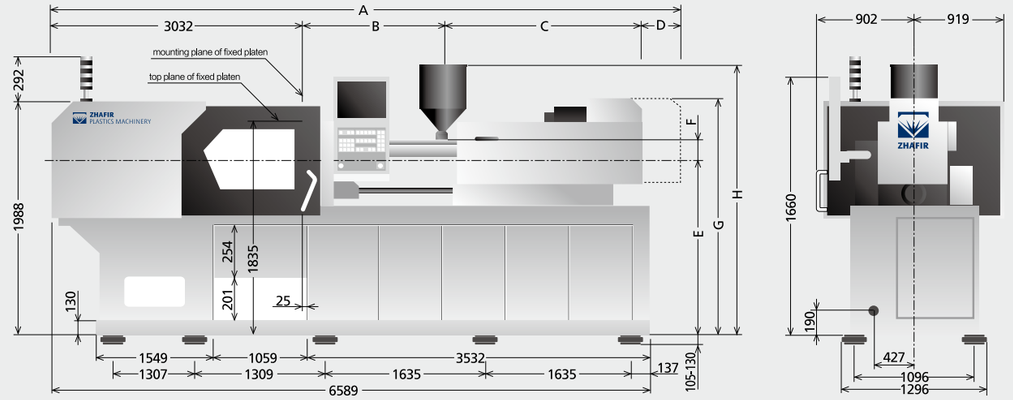
		A	B	C (OP)	A	B	C (OP)
Screw diameter	mm	45	50	55	50	55	60
Screw L/D ratio	L/D	25	22.5	23	25	22.7	23
Injection volume (theoretical)	cm^3	334	412	499	471	570	678
Injection weight (PS)	g	304	375	454	428	518	617
Screw speed	rpm		320			320	
Plasticizing capacity (PS)	g/s	35	46	60	46	60	71
Nozzle contact force	kN (tf)		54 (5.5)			54 (5.5)	
Heating power	kW	21.2	21.2	23.4	26.1	26.1	28.8
INJECTION UNIT			**830pns**			**1100pns**	
Injection speed	mm/s		160			160	
Injection rate (PS)	g/s	231	285	346	285	346	411
Injection pressure	Mpa	247	200	165	218	180	151
	bar	2470	2000	1650	2180	1800	1510
INJECTION UNIT			**830p**			**1100p**	
Injection speed	mm/s		300			300	
Injection rate (PS)	g/s	434	536	648	536	648	770
Injection pressure	Mpa	247	200	165	230	190	159
	bar	2470	2000	1650	2300	1900	1590

OTHERS			
Connection power	kW/A	830pns:47.5/85 830p:68.6/123	1100pns:51/92 1100p:78.6/139
Hopper capacity	L	50	50
Machine dimension	m	6.74×1.82×2.3	7.06×1.82×2.3
Machine weight	t	13.2	13.8

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE2300 Ⅱ /p

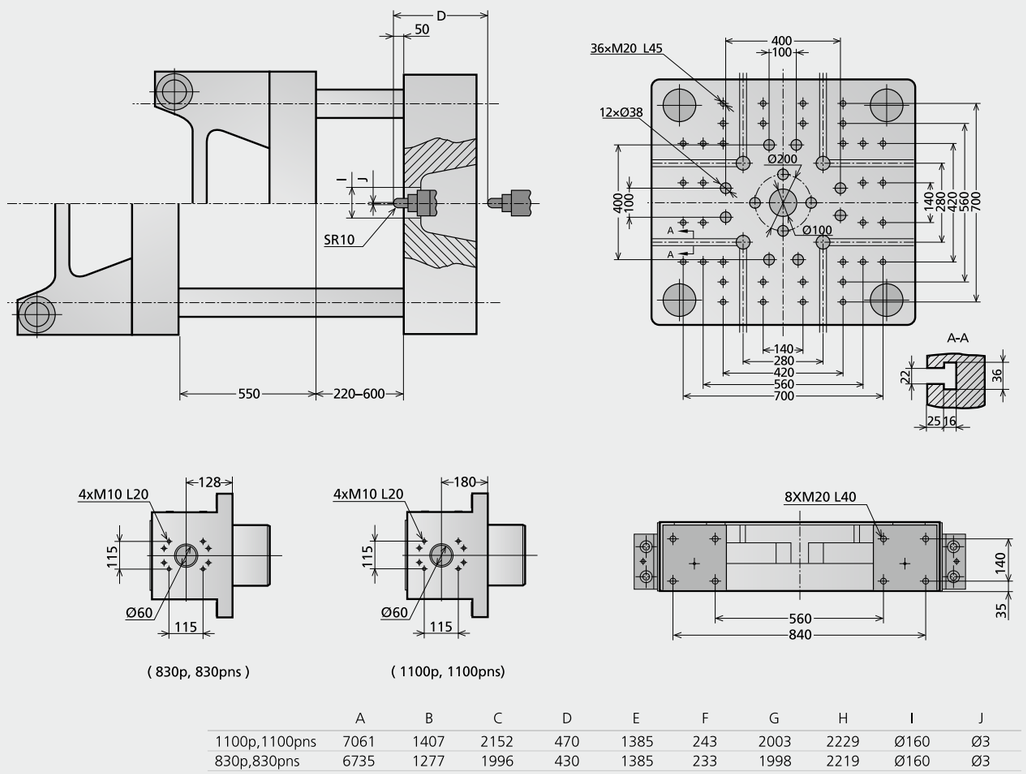
unit: mm



Note: The above machine size does not apply to the "C" type screw.

Platen dimensions VE2300 Ⅱ /p

unit: mm



(830p, 830pns)

(1100p, 1100pns)

	A	B	C	D	E	F	G	H	I	J
1100p,1100pns	7061	1407	2152	470	1385	243	2003	2229	Ø160	Ø3
830p,830pns	6735	1277	1996	430	1385	233	1998	2219	Ø160	Ø3

Technical data **VE3000 Ⅱ /p**

CLAMPING UNIT		VE3000 Ⅱ
Clamping force	kN	3000
Toggle stroke	mm	600
Min. mold height	mm	240
Max. mold height	mm	650
Max. mold opening stroke	mm	1250
Space between tie bars H×V	mm	670×620
Min. mold dimension	mm	435×400
Ejector stroke	mm	150
Ejector force	kN (tf)	58.8 (6.0)
Mold platen dimension H×V	mm	1000×950

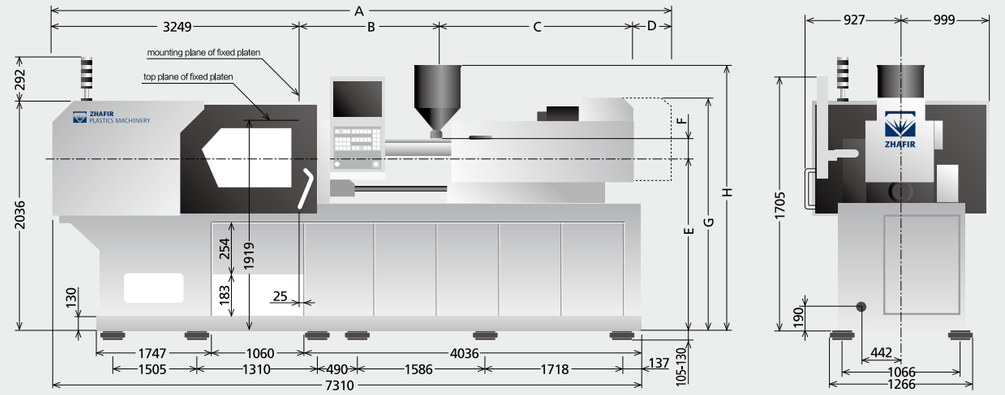
		A	B	C (OP)	A	B	C (OP)	A	B	C (OP)
Screw diameter	mm	50	55	60	55	60	65	60	65	70
Screw L/D ratio	L/D	25	22.7	23	25	22.9	23	25	23	23
Injection volume (theoretical) cm^3		471	570	678	617	735	863	792	929	1078
Injection weight (PS)	g	428	518	617	562	668	785	720	845	980
Screw speed	rpm	320			300			250		
Plasticizing capacity (PS)	g/s	46	60	71	56	70	81	60	70	85
Nozzle contact force	kN (tf)	54 (5.5)			54 (5.5)			54 (5.5)		
Heating power	kW	26.1	26.1	28.8	32.9	32.9	35.6	36.1	36.1	38.8
INJECTION UNIT		**1100pns**			**1400pns**			**1700pns**		
Injection speed	mm/s	160			160			160		
Injection rate (PS)	g/s	285	346	411	346	411	483	411	483	560
Injection pressure	Mpa	218	180	151	214	180	153	210	180	155
	bar	2180	1800	1510	2140	1800	1530	2100	1800	1550
INJECTION UNIT		**1100p**			**1400p**			**1700p**		
Injection speed	mm/s	300			300			300		
Injection rate (PS)	g/s	536	648	770	648	770	905	770	905	1050
Injection pressure	Mpa	230	190	159	230	193	164	230	196	169
	bar	2300	1900	1590	2300	1930	1640	2300	1960	1690

OTHERS				
Connection power	kW/A	1100pns:51/92 1100p:78.6/139	1400pns:56/101 1400p:89.2/157	1700pns:72/129 1700p:95.5/168
Hopper capacity	L	50	50	50
Machine dimension	m	7.36×1.93×2.33	7.76×1.93×2.33	7.95×1.93×2.33
Machine weight	t	16.3	17.2	17.6

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE 3000 Ⅱ /p

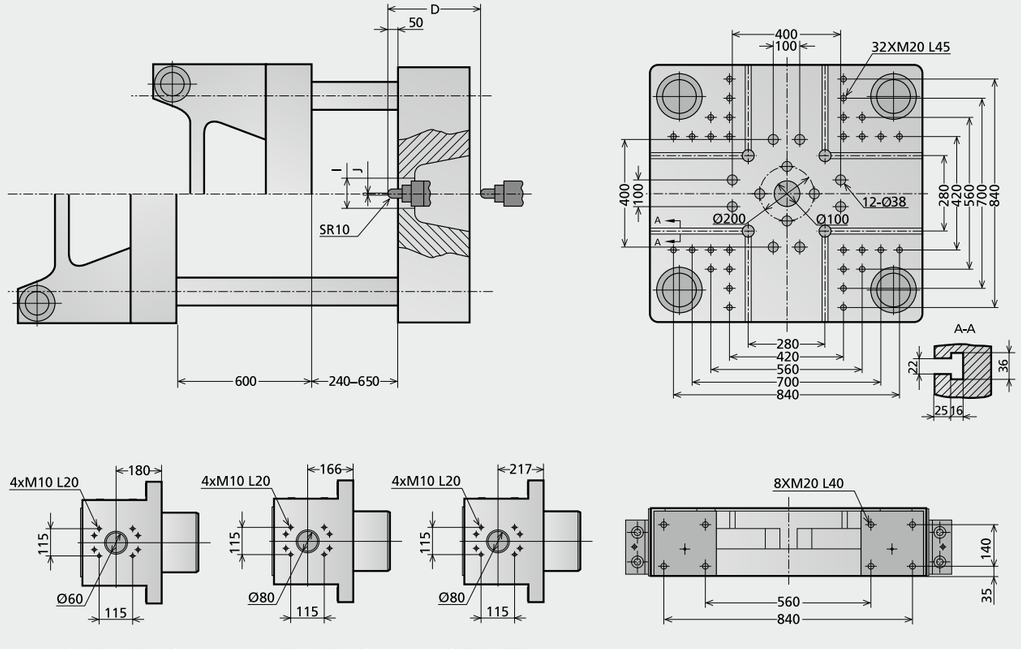
unit: mm



Note: The above machine size does not apply to the "C" type screw.

Platen dimensions VE3000 Ⅱ /p

unit: mm



	A	B	C	D	E	F	G	H	I	J
1700p,1700pns	7998	1697	2452	600	1388	248	2083	2237	Ø160	Ø3
1400p,1400pns	7752	1562	2401	540	1388	248	2083	2237	Ø160	Ø3
1100p,1100pns	7348	1407	2152	540	1388	243	2006	2232	Ø160	Ø3

Technical data **VE3800 Ⅱ /p**

CLAMPING UNIT		VE3800 Ⅱ
Clamping force	kN	3800
Toggle stroke	mm	660
Min. mold height	mm	250
Max. mold height	mm	710
Max. mold opening stroke	mm	1370
Space between tie bars H×V	mm	770×770
Min. mold dimension	mm	500×500
Ejector stroke	mm	160
Ejector force	kN (tf)	58.8 (6.0)
Mold platen dimension H×V	mm	1150×1150

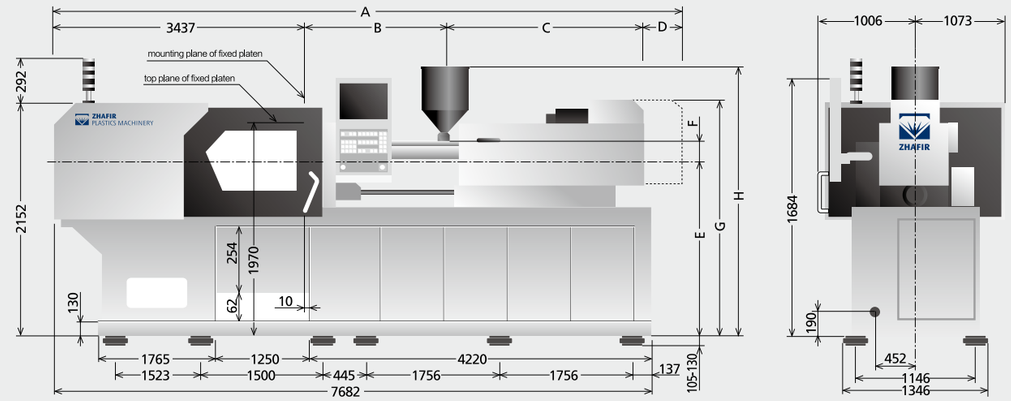
		A	B	C (OP)	A	B	C (OP)	A	B	C (OP)
Screw diameter	mm	50	55	60	55	60	65	60	65	70
Screw L/D ratio	L/D	25	22.7	23	25	22.9	23	25	23	23
Injection volume (theoretical)	cm^3	471	570	678	617	735	863	792	929	1078
Injection weight (PS)	g	428	518	617	562	668	785	720	845	980
Screw speed	rpm	320			300			250		
Plasticizing capacity (PS)	g/s	46	60	71	56	70	81	60	70	85
Nozzle contact force	kN (tf)	54 (5.5)			54 (5.5)			54 (5.5)		
Heating power	kW	26.1	26.1	28.8	32.9	32.9	35.6	36.1	36.1	38.8
INJECTION UNIT		**1100pns**			**1400pns**			**1700pns**		
Injection speed	mm/s	160			160			160		
Injection rate (PS)	g/s	285	346	411	346	411	483	411	483	560
Injection pressure	Mpa	218	180	151	214	180	153	210	180	155
	bar	2180	1800	1510	2140	1800	1530	2100	1800	1550
INJECTION UNIT		**1100p**			**1400p**			**1700p**		
Injection speed	mm/s	300			300			300		
Injection rate (PS)	g/s	536	648	770	648	770	905	770	905	1050
Injection pressure	Mpa	230	190	159	230	193	164	230	196	169
	bar	2300	1900	1590	2300	1930	1640	2300	1960	1690

OTHERS				
Connection power	kW/A	1100pns:51/92 1100p:78.6/139	1400pns:56/101 1400p:89.2/157	1700pns:72/129 1700p:95.5/168
Hopper capacity	L	50	50	50
Machine dimension	m	7.58×2.1×2.45	8.03×2.1×2.45	8.17×2.1×2.45
Machine weight	t	19.3	20.2	20.6

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE3800 Ⅱ /p

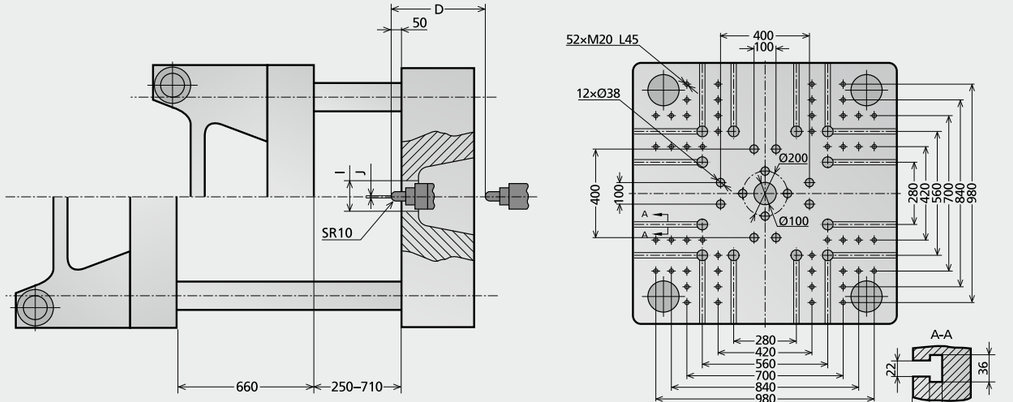
unit: mm

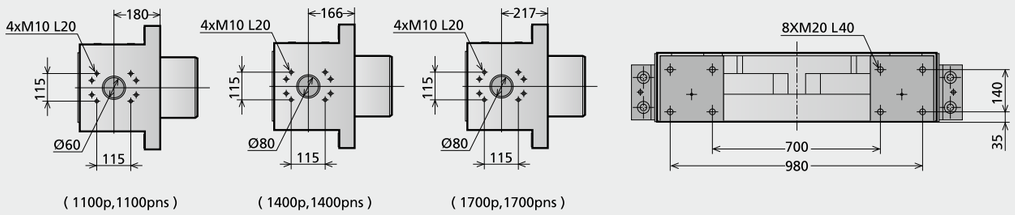


Note: The above machine size does not apply to the "C" type screw.

Platen dimensions VE3800 Ⅱ /p

unit: mm





(1100p,1100pns) (1400p,1400pns) (1700p,1700pns)

	A	B	C	D	E	F	G	H	I	J
1700p,1700pns	8186	1697	2452	600	1388	248	2083	2237	Ø160	Ø3
1400p,1400pns	8000	1562	2401	600	1388	248	2083	2237	Ø160	Ø3
1100p,1100pns	7576	1407	2152	580	1388	243	2006	2232	Ø160	Ø3

Technical data **VE4500 Ⅱ /p**

CLAMPING UNIT		VE4500 Ⅱ
Clamping force	kN	4500
Toggle stroke	mm	800
Min. mold height	mm	400
Max. mold height	mm	820
Max. mold opening stroke	mm	1620
Space between tie bars H×V	mm	820×820
Min. mold dimension	mm	540×540
Ejector stroke	mm	180
Ejector force	kN (tf)	117.6(12.0)
Mold platen dimension H×V	mm	1220×1220

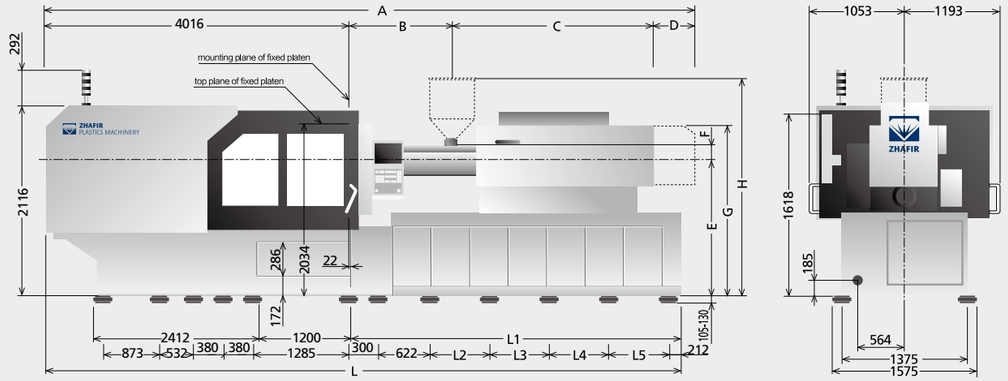
		A	B	C(OP)	A	B	C(OP)
Screw diameter	mm	55	60	65	60	65	70
Screw L/D ratio	L/D	25	22.9	23	25	23	23
Injection volume (theoretical)	cm^3	617	735	863	792	929	1078
Injection weight (PS)	g	562	668	785	720	845	980
Screw speed	rpm		300			250	
Plasticizing capacity (PS)	g/s	56	70	81	60	70	85
Nozzle contact force	kN (tf)		54(5.5)			54(5.5)	
Heating power	kW	32.9	32.9	35.6	36.1	36.1	38.8
INJECTION UNIT			**1400pns**			**1700pns**	
Injection speed	mm/s		160			160	
Injection rate (PS)	g/s	346	411	483	411	483	560
Injection pressure	Mpa	214	180	153	210	180	155
	bar	2140	1800	1530	2100	1800	1550
INJECTION UNIT			**1400p**			**1700p**	
Injection speed	mm/s		300			300	
Injection rate (PS)	g/s	648	770	905	770	905	1050
Injection pressure	Mpa	230	193	164	230	196	169
	bar	2300	1930	1640	2300	1960	1690

OTHERS			
Connection power	kW/A	1400pns:56/101 1400p:89.2/157	1700pns:72/129 1700p:95.5/168
Hopper capacity	L	50	50
Machine dimension	m	8.8*2.3*2.42	8.8*2.3*2.42
Machine weight	t	25	25.4

We reserve the right to make changes as a result of further technical advantages.

Machine dimensions VE4500 Ⅱ /p

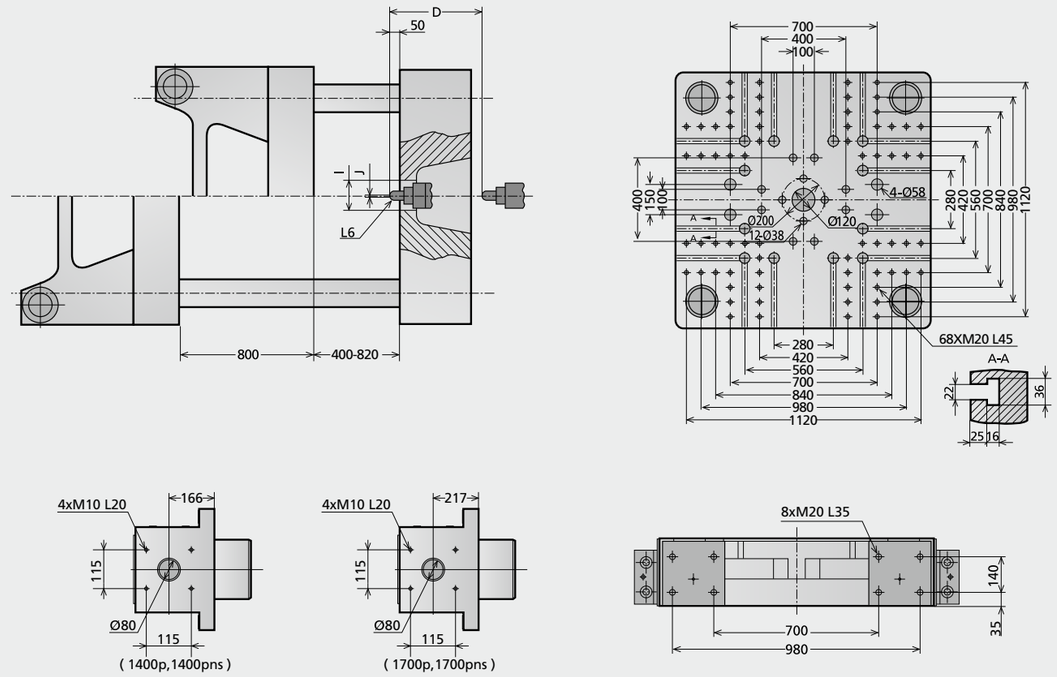
unit: mm



Note: The above machine size does not apply to the "C" type screw.

Platen dimensions VE4500 Ⅱ /p

unit: mm



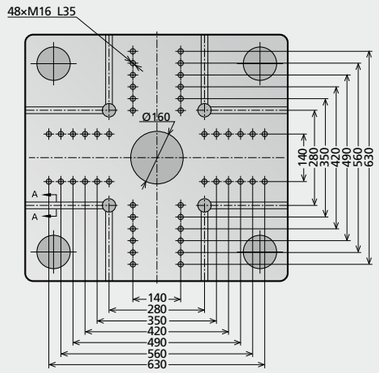
	A	B	C	D	E	F	G	H	I	J	L	L1	L2	L3	L4	L5	L6
1700p,1700pns	8765	1697	2452	600	1404	248	2100	2233	Ø200	Ø3	8785	4747	888	950	904	844	SR10
1400p,1400pns	8579	1562	2401	600	1404	248	2100	2233	Ø200	Ø3	8785	4747	888	950	904	844	SR10

Platen Layout **VE1500 Ⅱ /p**

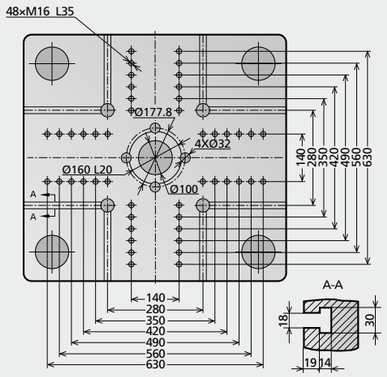
European version

unit: mm

Fixed platen



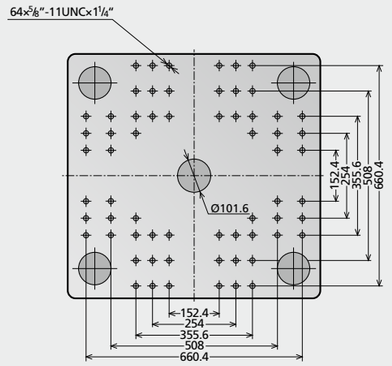
Movable platen



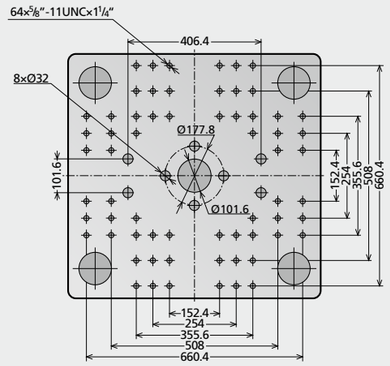
American version

unit: mm

Fixed platen



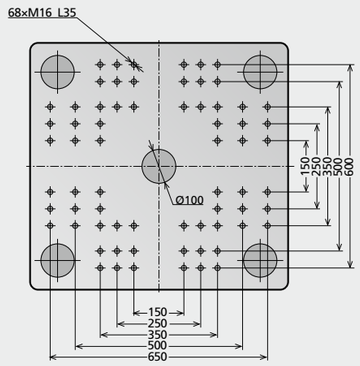
Movable platen



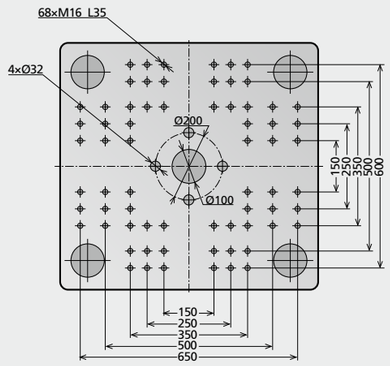
Japanese version

unit: mm

Fixed platen



Movable platen

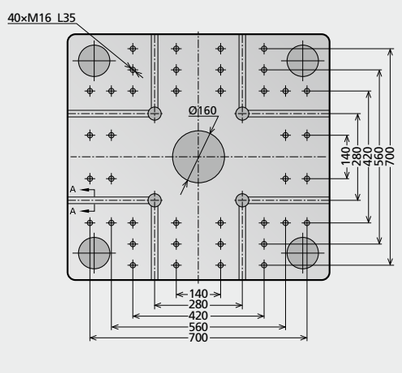


Platen Layout **VE1900 Ⅱ /p**

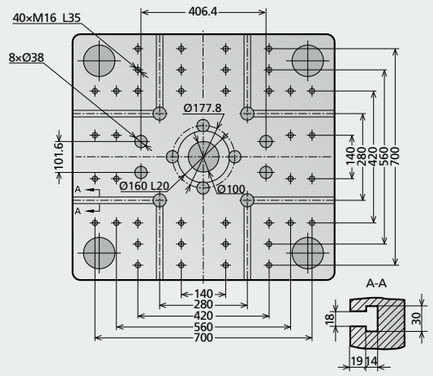
European version

unit: mm

Fixed platen



Movable platen



American version

unit: mm

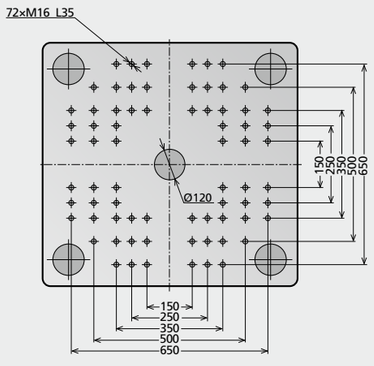
Fixed platen

Movable platen

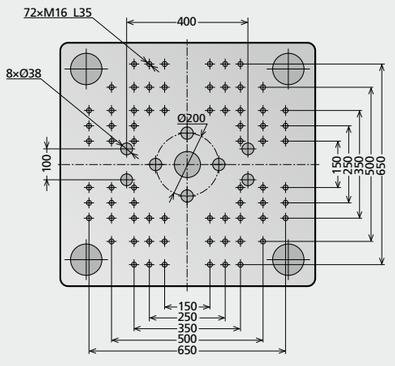
Japanese version

unit: mm

Fixed platen



Movable platen

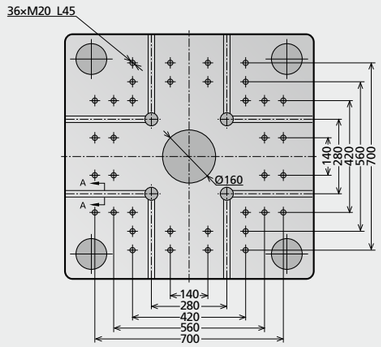


Platen Layout **VE2300 Ⅱ /p**

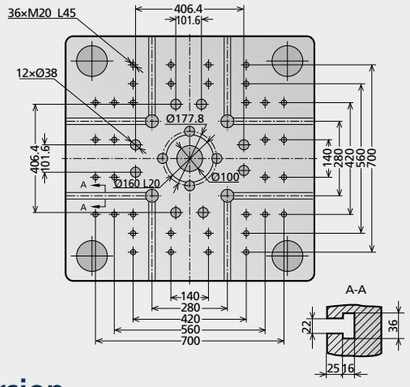
European version

unit: mm

Fixed platen



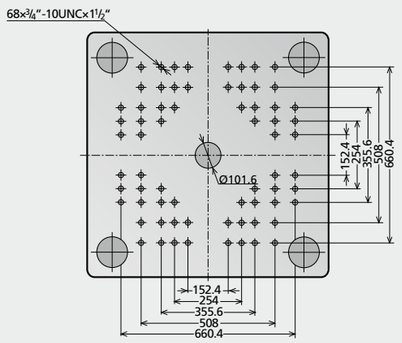
Movable platen



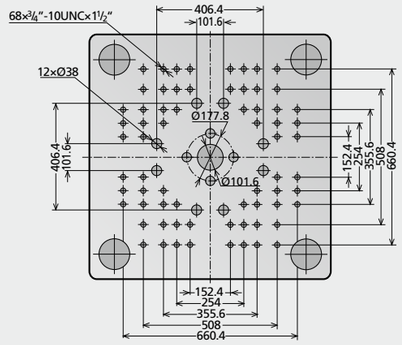
American version

unit: mm

Fixed platen



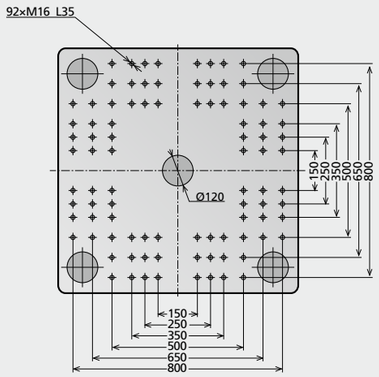
Movable platen



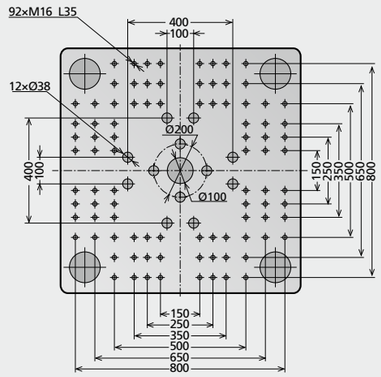
Japanese version

unit: mm

Fixed platen



Movable platen

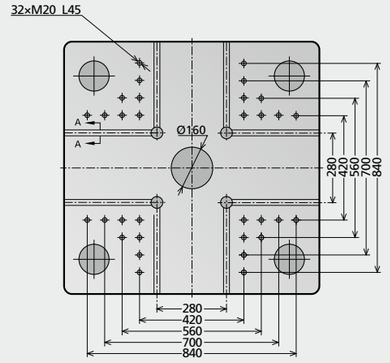


Platen Layout **VE3000 Ⅱ /p**

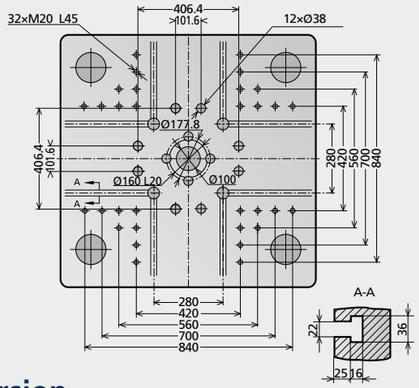
European version

unit: mm

Fixed platen



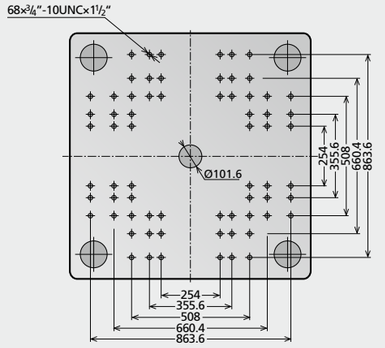
Movable platen



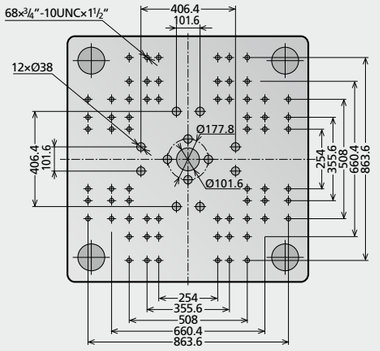
American version

unit: mm

Fixed platen



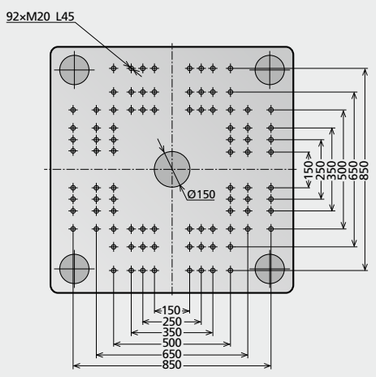
Movable platen



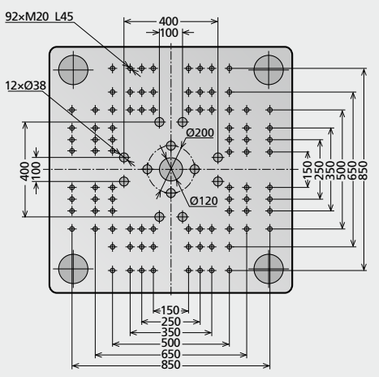
Japanese version

unit: mm

Fixed platen



Movable platen

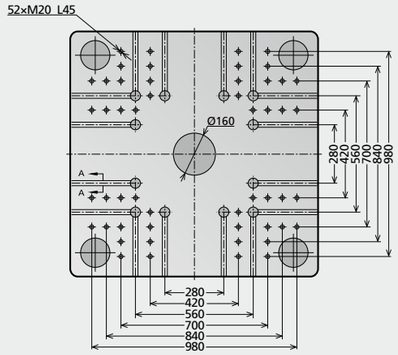


Platen Layout **VE3800 Ⅱ /p**

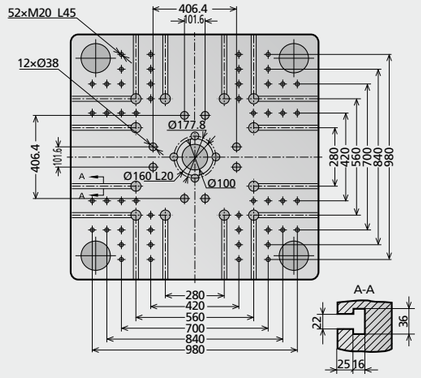
European version

unit: mm

Fixed platen



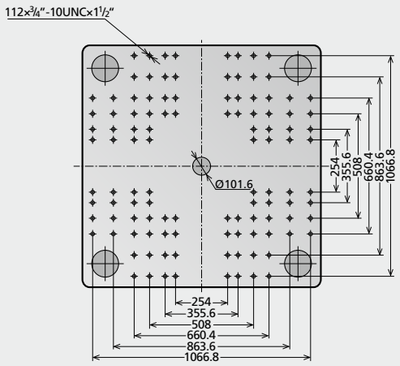
Movable platen



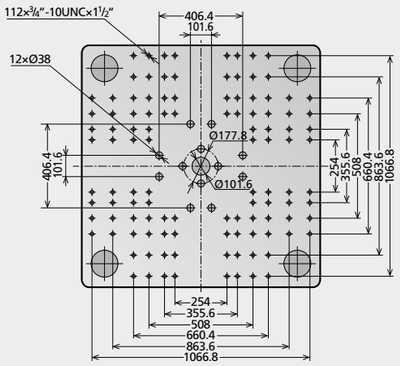
American version

unit: mm

Fixed platen



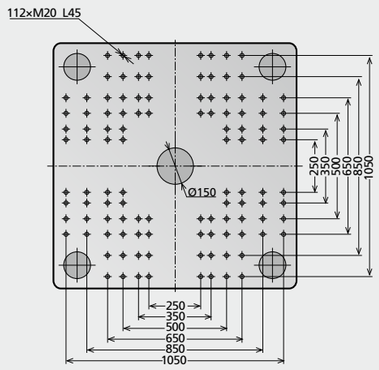
Movable platen



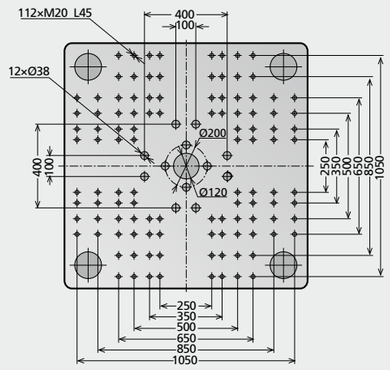
Japanese version

unit: mm

Fixed platen



Movable platen

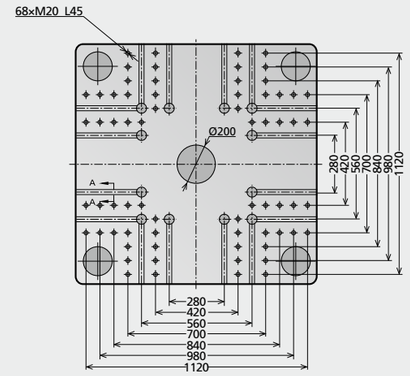


Platen Layout **VE4500 Ⅱ /p**

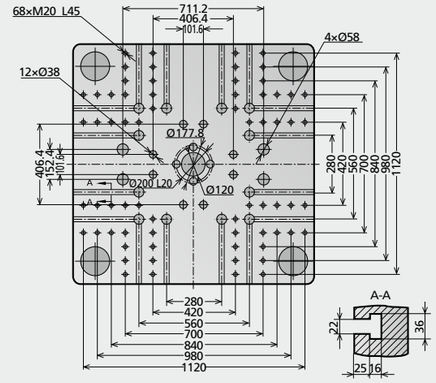
European version

unit: mm

Fixed platen



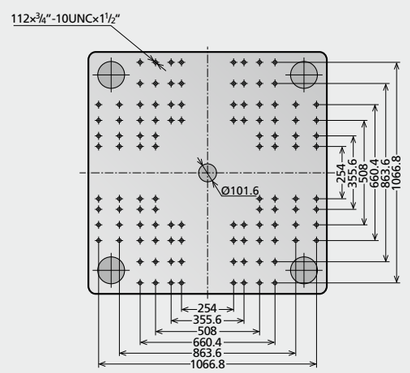
Movable platen



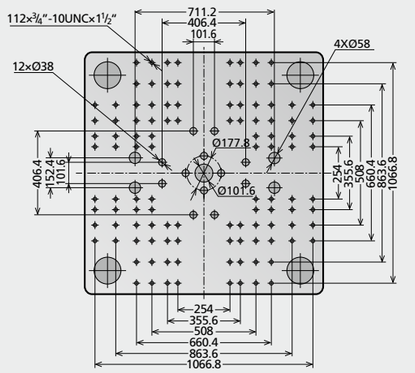
American version

unit: mm

Fixed platen



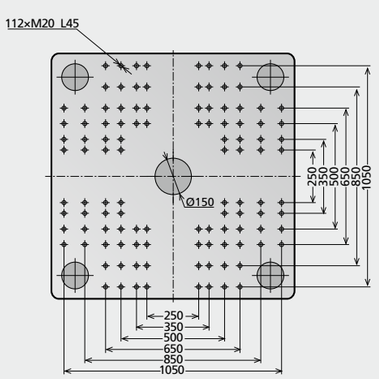
Movable platen



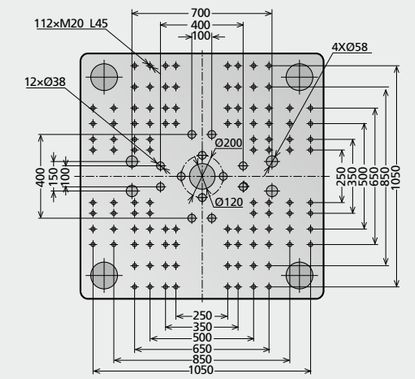
Japanese version

unit: mm

Fixed platen



Movable platen



Standard Equipment List

Injection Unit

- Abrasion-resistant screw unit(open nozzle)
- Nozzle safety guard
- 6 Injection steps
- 4 Pressure holding steps
- 3 Dosing steps
- 3 Back pressure steps
- Screw suck back select(After holding/After dosing)
- Screw suck back delay function
- Injection delay function
- Dosing delay function
- Intrusion/filling by rotating screw
- V/P switch over mode(position,time,pressure,speed)
- Injection speed response mode select
- Injection pressure segment control
- Speed limit during holding pressure
- Screw postion setting(unit: 0.01mm)
- Screw rotation speed setting
- Holding pressure time setting
- Mold open available during dosing
- Barrel heating closed-loop control (K/J type available)
- Barrel temperature holding function
- Barrel temperature auto tunning function
- Barrel pre-heating function
- Barrel temperature synchronous control
- Material overheating prevention function
- Screw cold start prevention
- Auto purge function
- Screw rotation speed display
- Setting for nozzle movement (switch/time)
- Selection of nozzle retract (3 modes)
- Nozzle contact device
- Nozzle centre adjustable
- Feeding throat temperature closed-loop control
- Injection unit swiveling device
- Independent nozzle temperature control

Clamping unit

- 6 mold moving steps
- Mold safety protection
- Mold setup mode
- Injection compression function
- Clamping force pre-release during cooling
- Injection during clamping(Pre-injection)
- 2 Step mold close/clamp process
- In mold cut function (ejector activate when mold closed)
- Clamping safety device(Mechanical and Electrical)
- Adjustable moving platen support
- Clamping force setting
- Ejector device
- Ejector function deactivatable
- 3 Ejector steps
- Ejector forward delay function
- Ejector vibration function
- Ejector A/B function(2 steps ejector stroke)
- Eject on the fly

Clamping unit

- Ejector plate return signal confirmation
- Cooling water distributor (12 sets up to 1500Ⅱ, 18 sets for others)
- 4 air valve groups
- Embedded locating ring (fixed platen)
- In mold ejector pin protection interface
- Mold installation mode (Low speed mold move)
- Needle shut off nozzle interface (one spare electrical interface)
- Emergency stop(Operating and Non-operating sides)
- Robot mounting layout
- Central lubrication system
- S-curve flexible control for mold movement

Controller & Monitor

- 15 inch TFT color touch screen
- Mold profile data memory (up to 200 sets)
- Alarm history
- Data setting record
- 3 USB R/W interface
- Injection pressure and speed curve display and record
- Euromap 12 robot interface
- Multi-language available(Chinese, German, English, Japanese etc.)
- Metric/Imperial unit select
- I/O monitor display
- Printer interface(USB connector)
- Production cycle monitor
- New page for production
- Production data record (5000 cycles display, 100,000 cycles record)
- Production data graphics
- Quality parameters tolerance setting
- Quality abnormal alarm
- Cycle counter
- Machine overview display
- Main data setting fast access
- Machine maintenance administration
- Clamping force curve display
- Barrel temperature monitor
- Display cycle time sequences in each phase
- 3 Color alarm lamp(red/yellow/green)
- Alarm buzzer
- HPM function: Auto limit injection pressure
- Display of actual performance data
- Multi-action selectable during machine alarm

Others

- Standard Zhafir color
- Adjustable machine pads
- 3 Power sockets (two 16A and one 32A)
- Standard hopper
- Mold clamper
- T-screw nuts(for T-slot use)
- Hopper slide movement support
- Tools box
- Spare parts
- Machine cover

Options List

Injection Unit

- Chrome-plating screw unit
- Corrosion and abrasion resistant screw unit
- High performance nozzle heating band
- Extended nozzle
- V/P switch based on in-mold pressure
- Needle shut off nozzle
- Spring nozzle
- Nozzle head customization
- Stainless steel hopper

Clamping unit

- Air blow
- Pneumatic ejector
- Hydraulic power pack(For core and valve gate)
- Hydraulic core interface (Programable)
- Pneumatic core interface (Programable)
- Valve Gate control interface
- Glass tube water flowmeter
- Customized mold platen(T-slot,threaded holes)
- Mold unscrew electrical interface
- Mold height extension
- Mold slide block protection interface
- Eject compression function
- Mold auto clamp device (Pneumatic/hydraulic)
- Ejector unit with brake
- Intergrated hot runner controller
- Product drop plate
- Automatic safety gate on operator side
- Clamping force automatically controller
- Quality flap
- Product drop detector

Controller & Monitor

- Mold temperature display and control
- Quality flap interface
- Heating disconnect detection function
- Additional cooling water distributor
- External transformer
- Euromap 67 robot interface
- Gas-assisted injection interface
- Magnetic platen interface
- Mold cavity pressure detection interface

Others

- Safty certification: CE,UL
- Infrared barrel heating band
- Barrel heating insulation cover

NOTE: